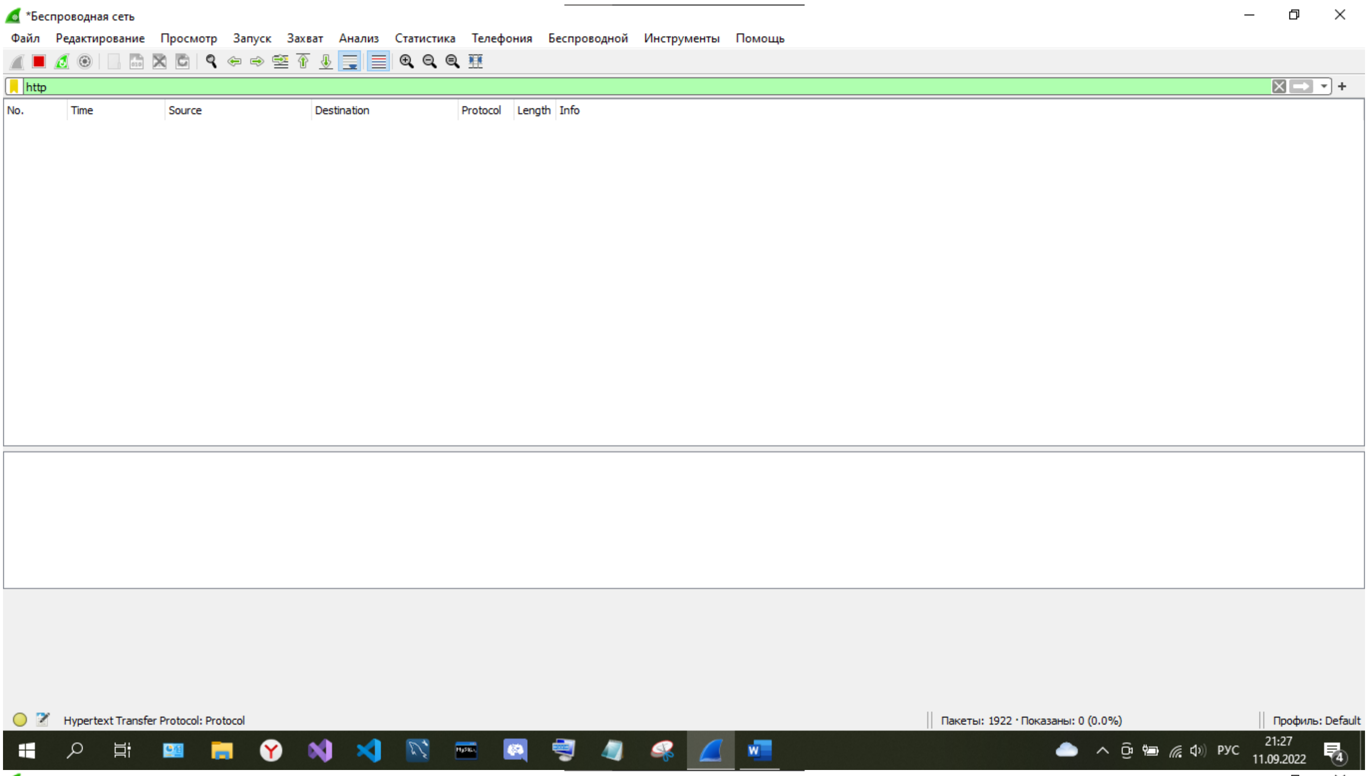
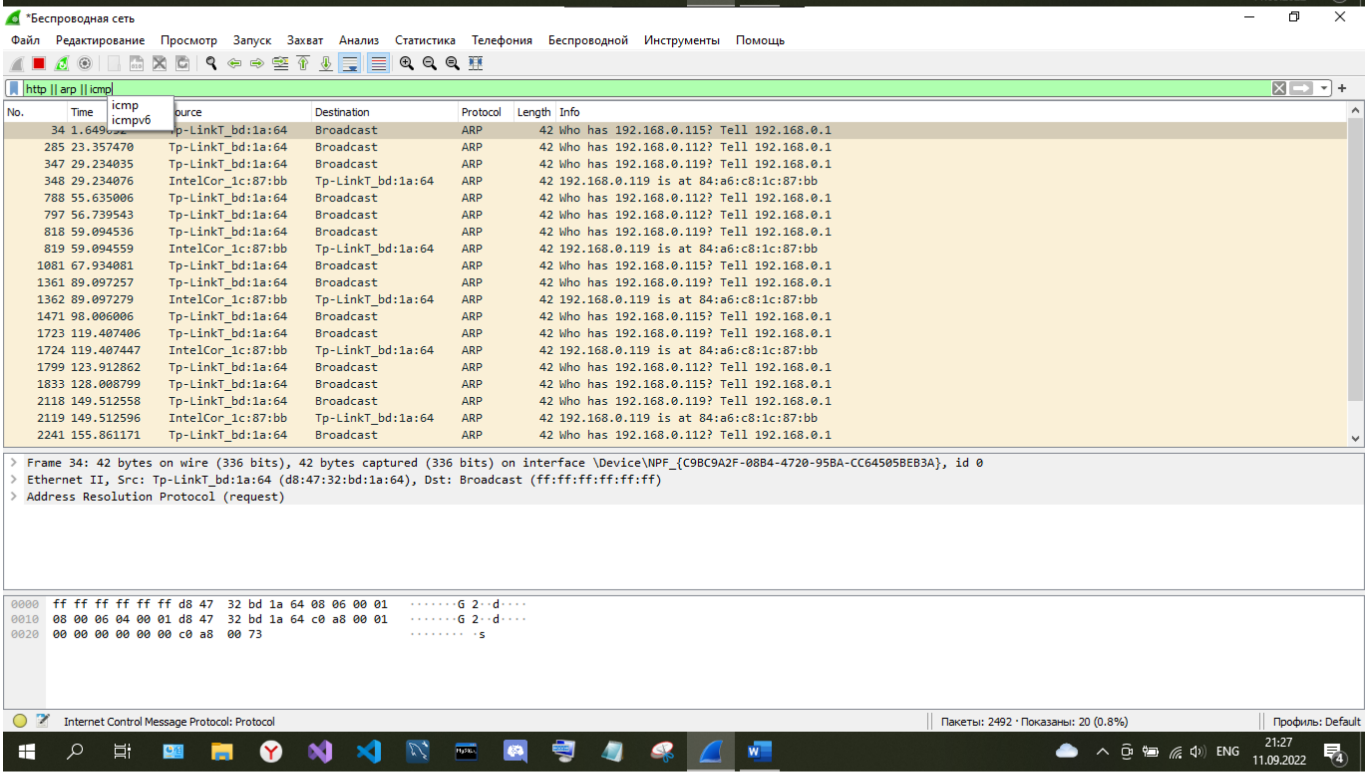
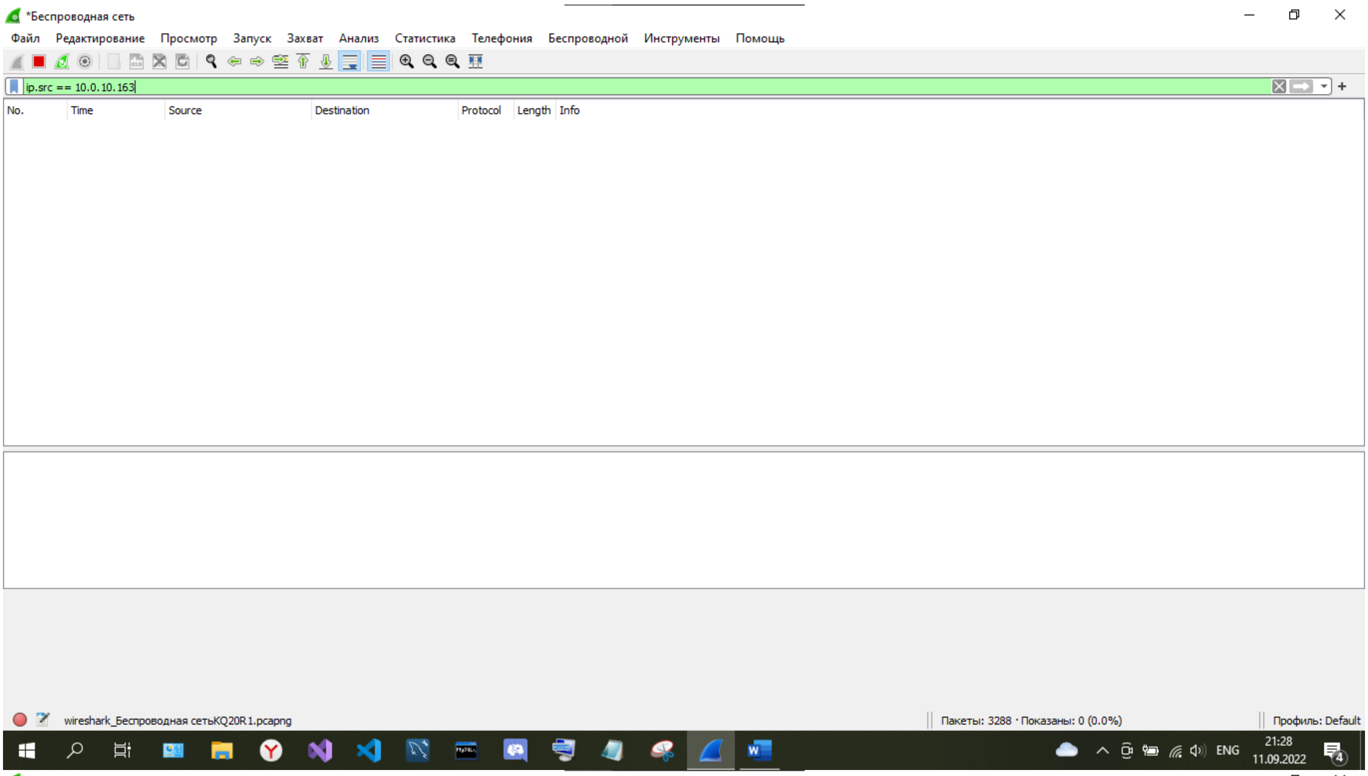
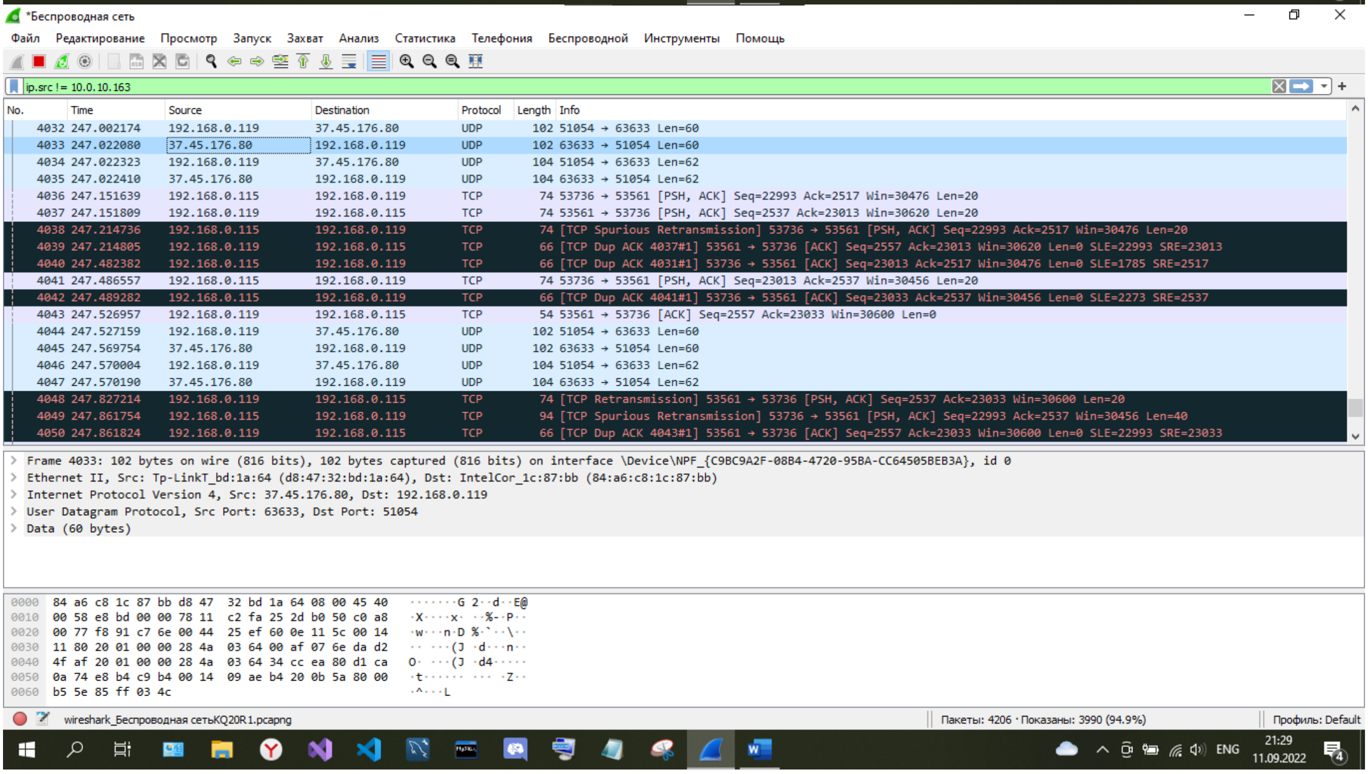
Цель работы: выработать навык анализа сетевого трафика.

Выполнение заданий:
Вариант 11

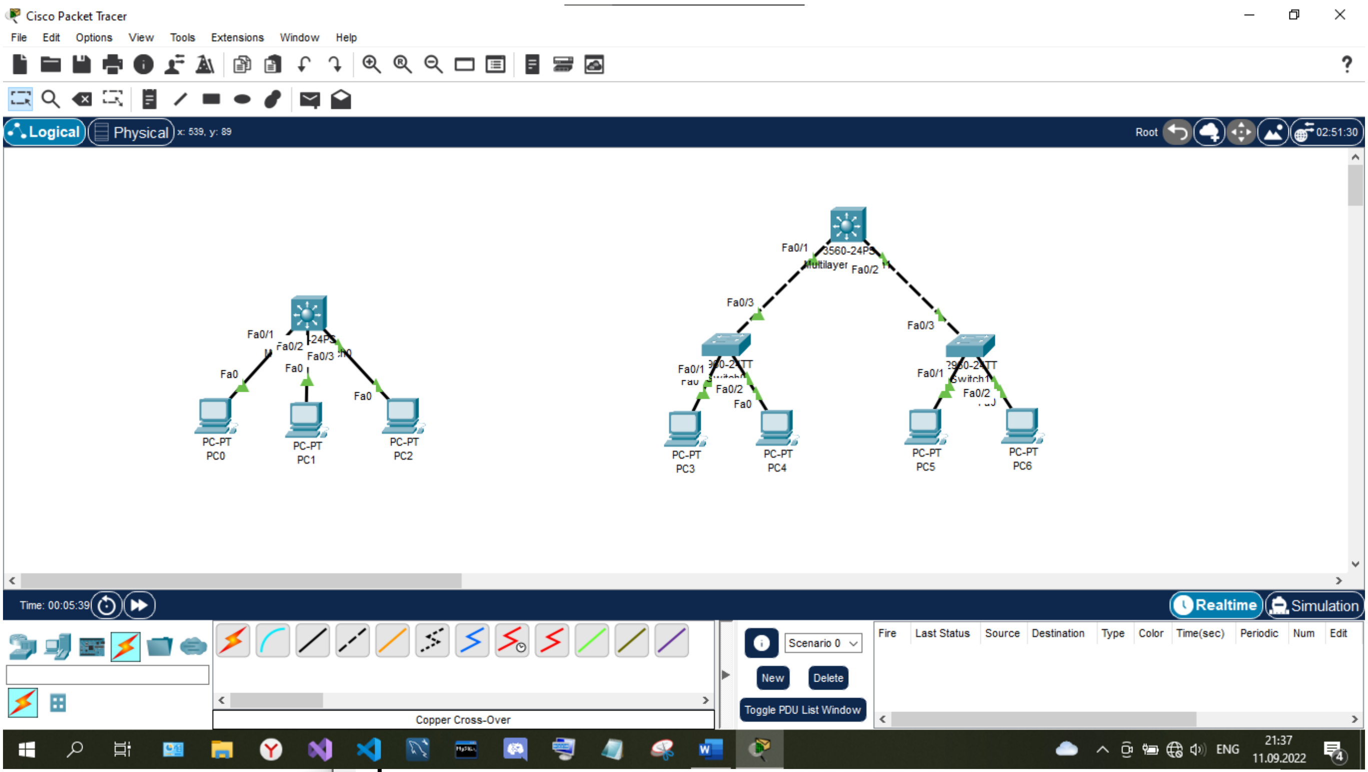


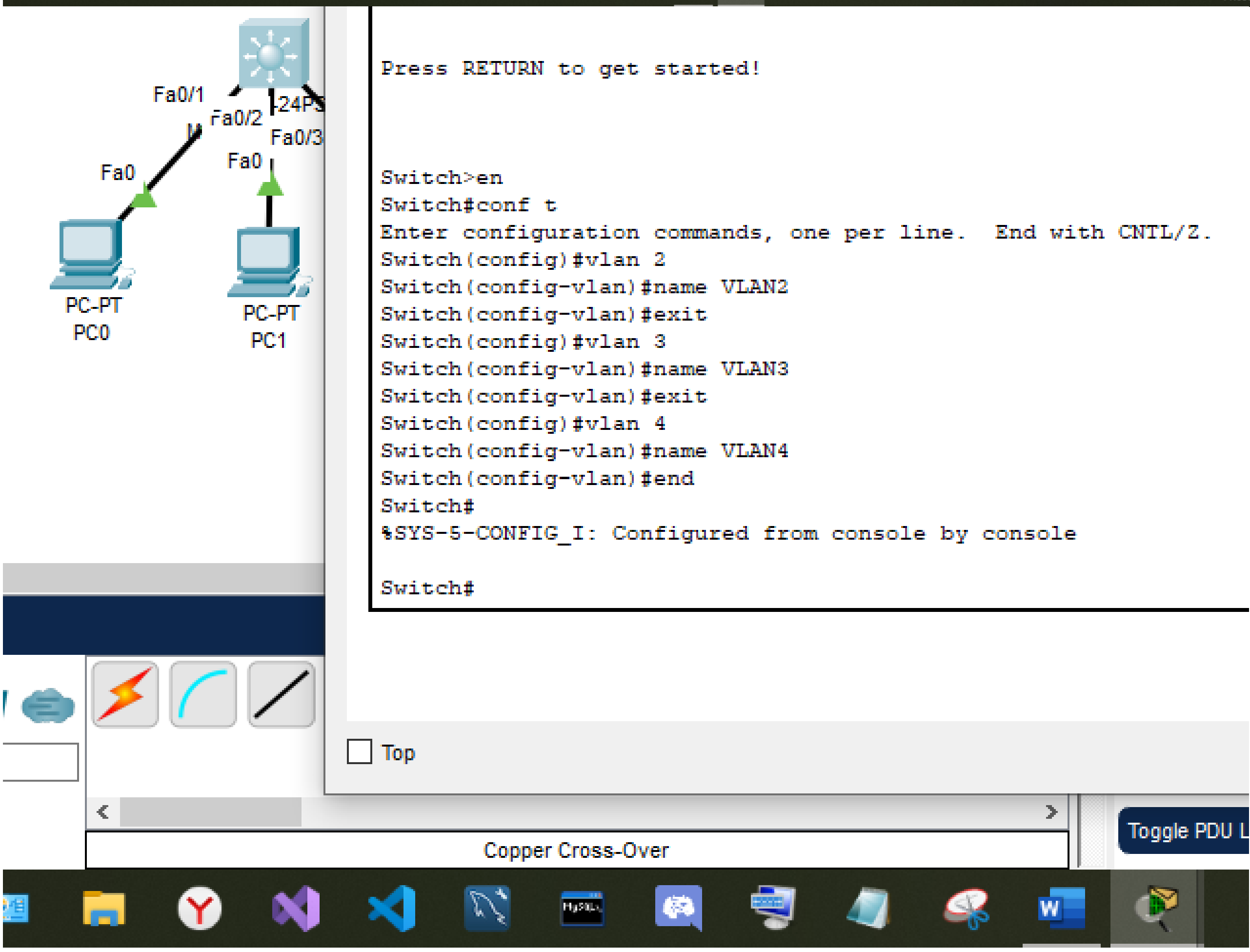


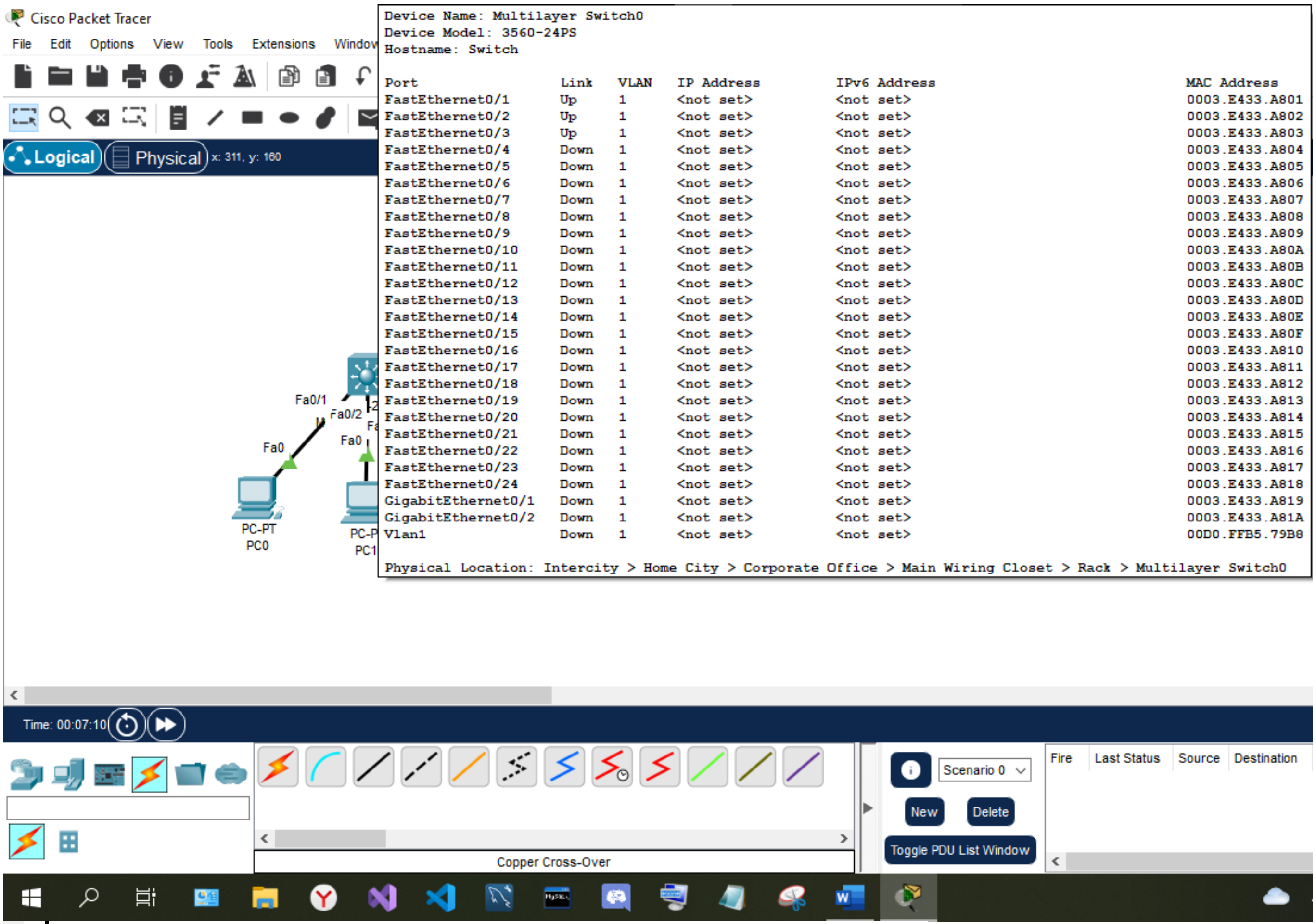




Layer 3 Switch


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 539, y: 89
Root
02:51:30
Fa0/1
Fa0/2
Fa0/3
Fa0
PC-PT
PC0
PC1
PC2
PC3
PC4
PC5
PC6
3560-24PS
Multilayer
Time: 00:05:39
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Copper Cross-Over
ENG
21:37
11.09.2022

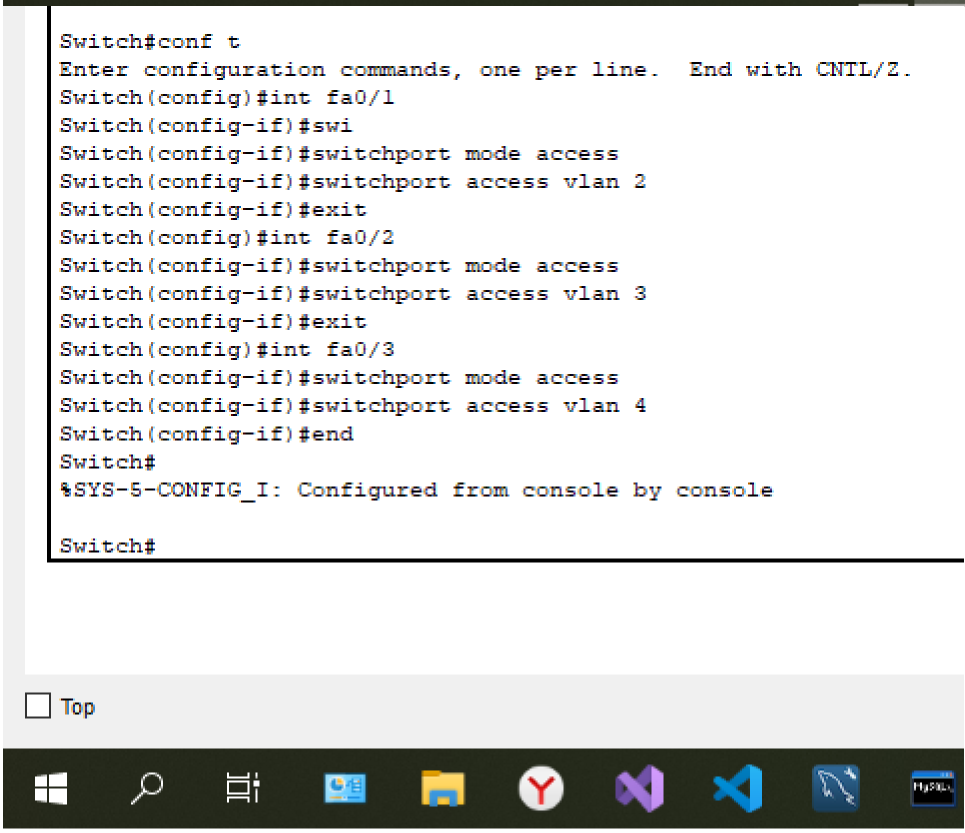

Fa0/1
Fa0/2
Fa0/3
Fa0
PC-PT
PC0
PC1
Press RETURN to get started!
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
Top
Copper Cross-Over
Toggle PDU L



Device Name: Multilayer Switch0
Device Model: 3560-24PS
Hostname: Switch

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Up	1	<not set>	<not set>	0003.E433.A801
FastEthernet0/2	Up	1	<not set>	<not set>	0003.E433.A802
FastEthernet0/3	Up	1	<not set>	<not set>	0003.E433.A803
FastEthernet0/4	Down	1	<not set>	<not set>	0003.E433.A804
FastEthernet0/5	Down	1	<not set>	<not set>	0003.E433.A805
FastEthernet0/6	Down	1	<not set>	<not set>	0003.E433.A806
FastEthernet0/7	Down	1	<not set>	<not set>	0003.E433.A807
FastEthernet0/8	Down	1	<not set>	<not set>	0003.E433.A808
FastEthernet0/9	Down	1	<not set>	<not set>	0003.E433.A809
FastEthernet0/10	Down	1	<not set>	<not set>	0003.E433.A80A
FastEthernet0/11	Down	1	<not set>	<not set>	0003.E433.A80B
FastEthernet0/12	Down	1	<not set>	<not set>	0003.E433.A80C
FastEthernet0/13	Down	1	<not set>	<not set>	0003.E433.A80D
FastEthernet0/14	Down	1	<not set>	<not set>	0003.E433.A80E
FastEthernet0/15	Down	1	<not set>	<not set>	0003.E433.A80F
FastEthernet0/16	Down	1	<not set>	<not set>	0003.E433.A810
FastEthernet0/17	Down	1	<not set>	<not set>	0003.E433.A811
FastEthernet0/18	Down	1	<not set>	<not set>	0003.E433.A812
FastEthernet0/19	Down	1	<not set>	<not set>	0003.E433.A813
FastEthernet0/20	Down	1	<not set>	<not set>	0003.E433.A814
FastEthernet0/21	Down	1	<not set>	<not set>	0003.E433.A815
FastEthernet0/22	Down	1	<not set>	<not set>	0003.E433.A816
FastEthernet0/23	Down	1	<not set>	<not set>	0003.E433.A817
FastEthernet0/24	Down	1	<not set>	<not set>	0003.E433.A818
GigabitEthernet0/1	Down	1	<not set>	<not set>	0003.E433.A819
GigabitEthernet0/2	Down	1	<not set>	<not set>	0003.E433.A81A
Vlan1	Down	1	<not set>	<not set>	00D0.FFB5.79B8

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Multilayer Switch0



```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

```
!
interface FastEthernet0/1
 switchport access vlan 2
 switchport mode access
 switchport nonegotiate
!
interface FastEthernet0/2
 switchport access vlan 3
 switchport mode access
 switchport nonegotiate
!
interface FastEthernet0/3
 switchport access vlan 4
 switchport mode access
 switchport nonegotiate
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
 --More--
```

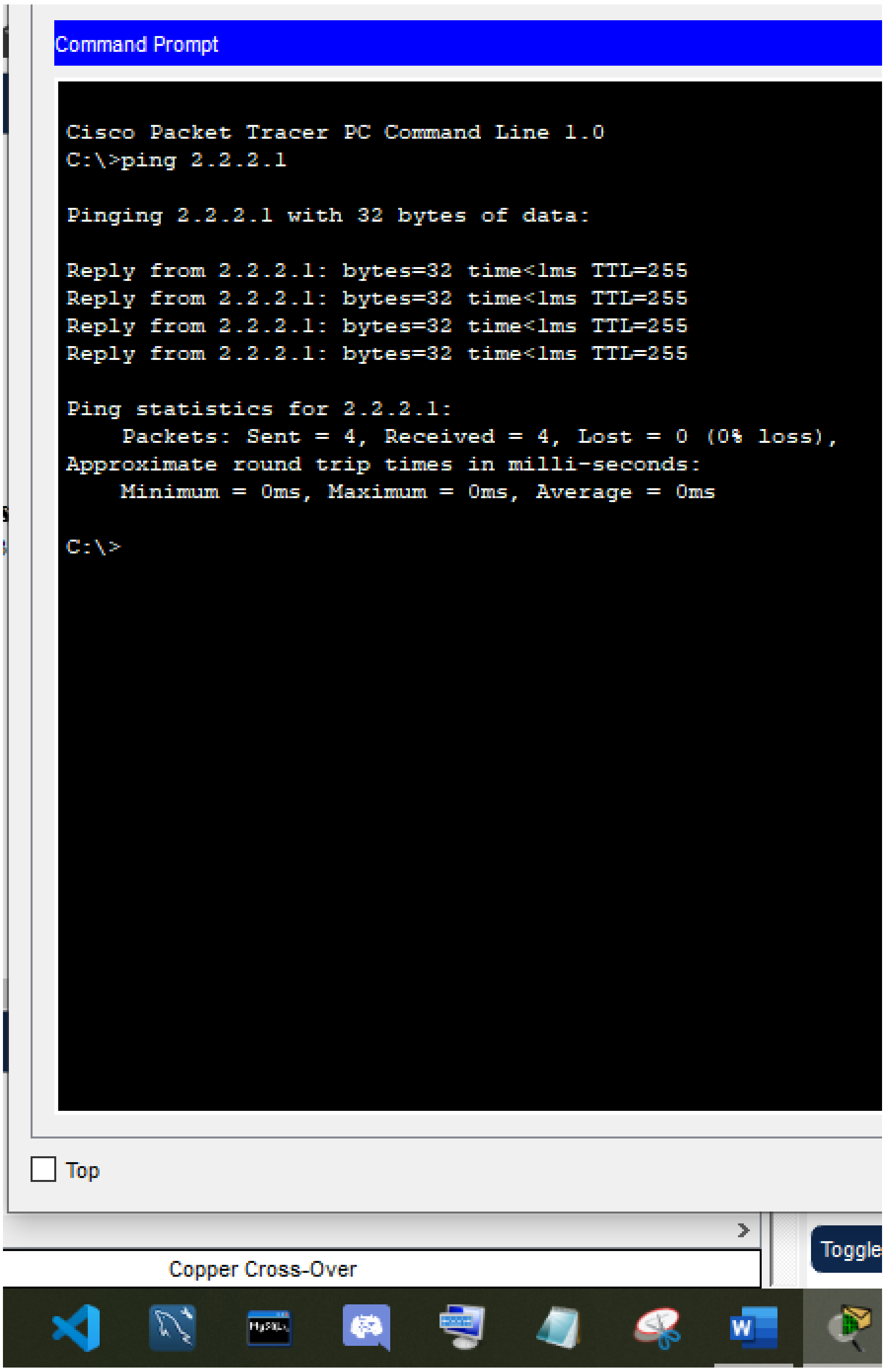
Top

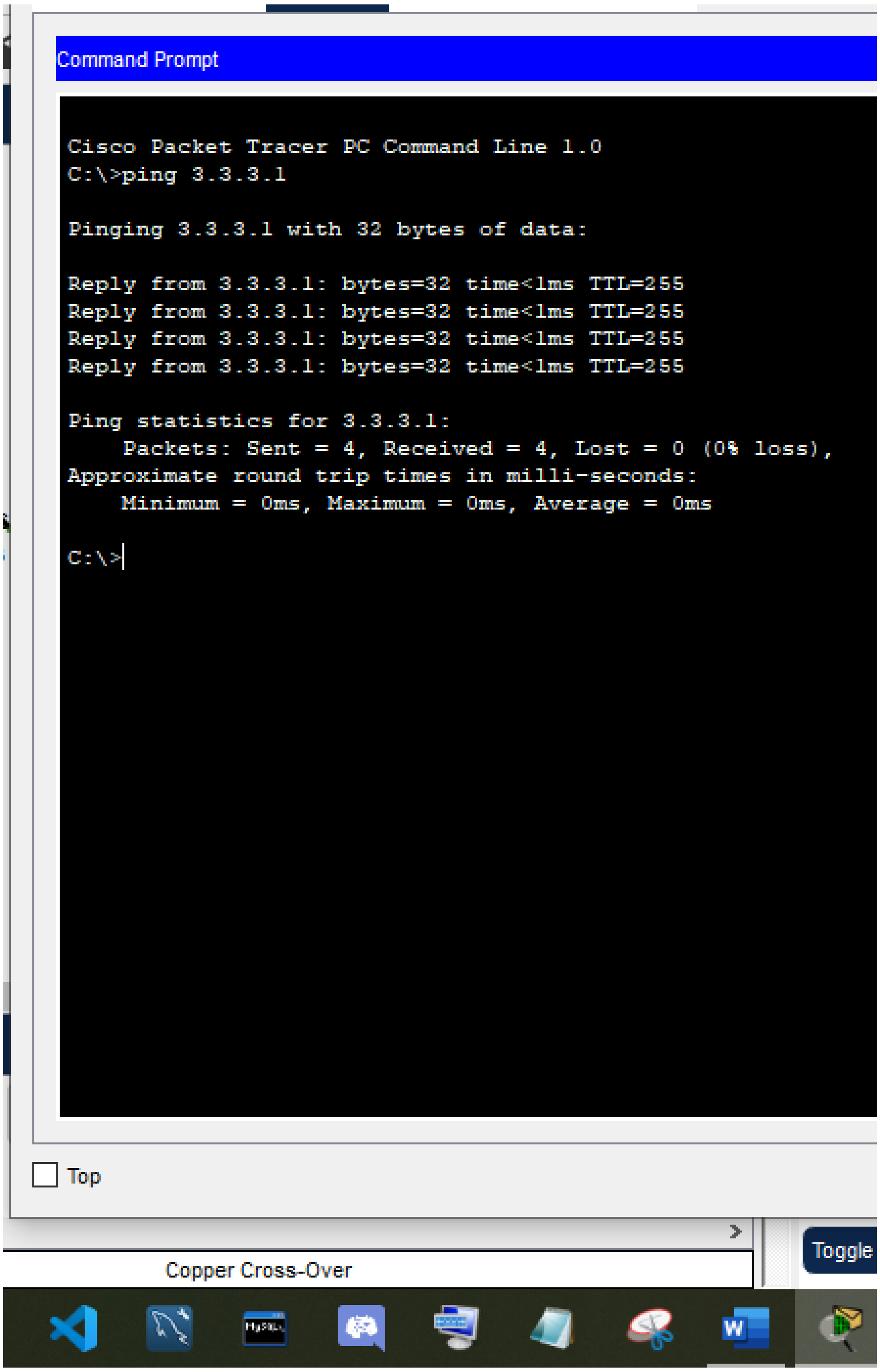
```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int vlan 2
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 3
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3, changed state to up

Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 4
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

Switch(config-if)#ip address 4.4.4.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

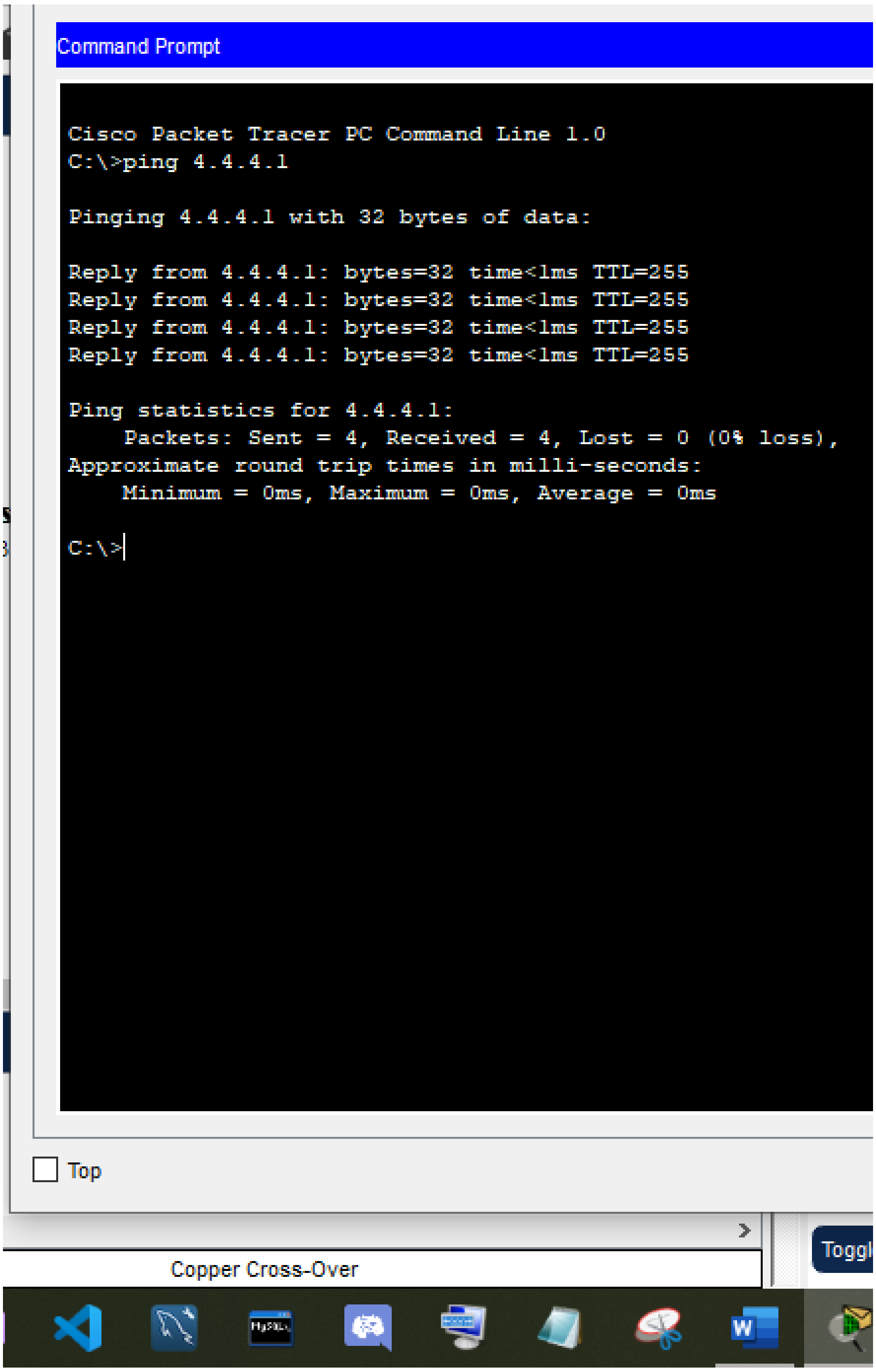
Top

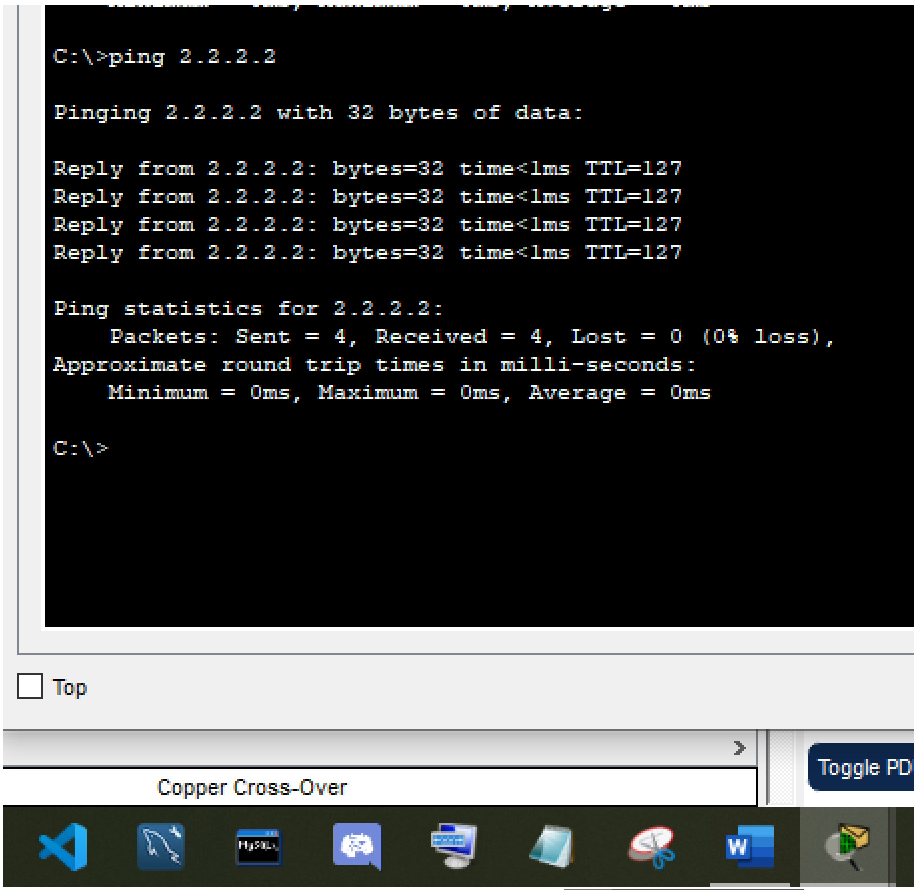
```
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan2
 mac-address 00d0.ffb5.7901
 ip address 2.2.2.1 255.255.255.0
!
interface Vlan3
 mac-address 00d0.ffb5.7902
 ip address 3.3.3.1 255.255.255.0
!
interface Vlan4
 mac-address 00d0.ffb5.7903
 ip address 4.4.4.1 255.255.255.0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
 --More--
```

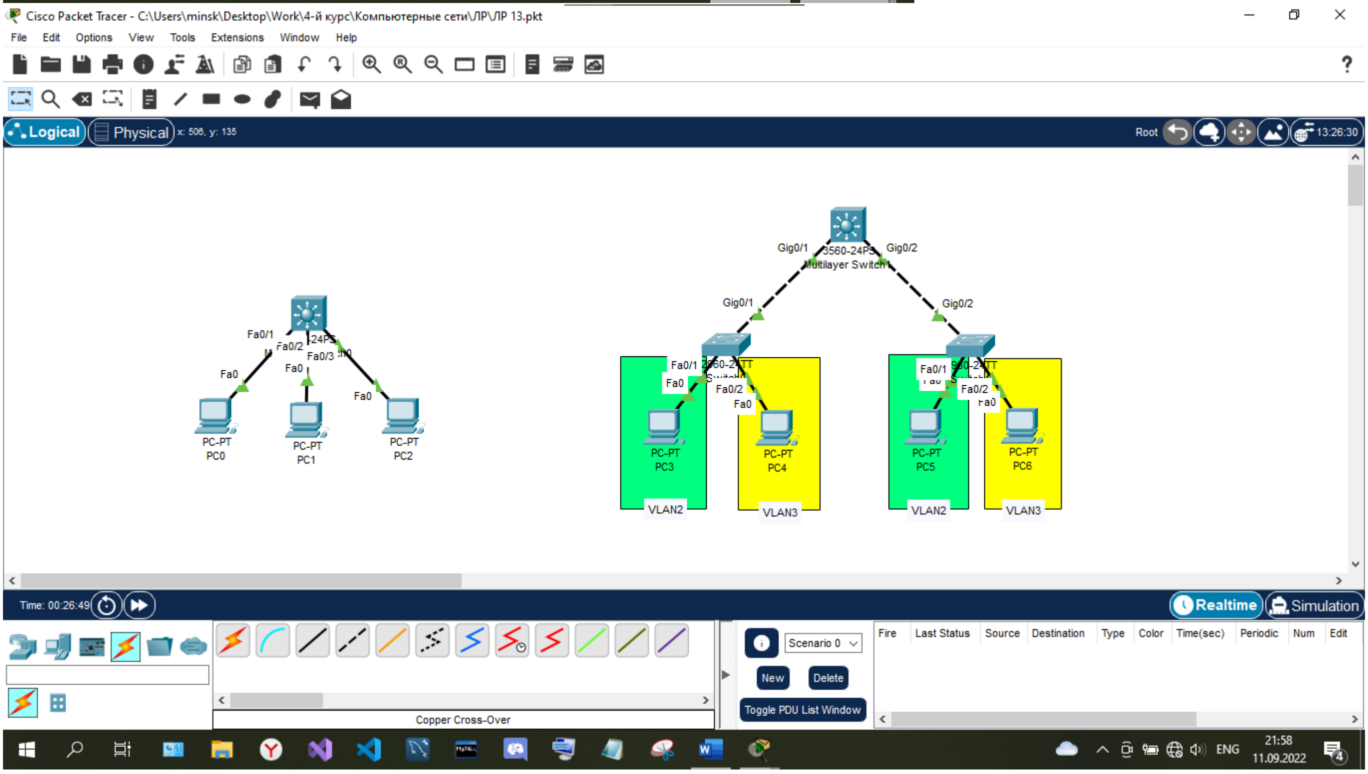
Top


Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2.2.2.1
Pinging 2.2.2.1 with 32 bytes of data:
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 2.2.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Copper Cross-Over
Toggle


Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.3.3.1
Pinging 3.3.3.1 with 32 bytes of data:
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Ping statistics for 3.3.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Copper Cross-Over
Toggle


Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 4.4.4.1
Pinging 4.4.4.1 with 32 bytes of data:
Reply from 4.4.4.1: bytes=32 time<1ms TTL=255
Reply from 4.4.4.1: bytes=32 time<1ms TTL=255
Reply from 4.4.4.1: bytes=32 time<1ms TTL=255
Reply from 4.4.4.1: bytes=32 time<1ms TTL=255
Ping statistics for 4.4.4.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Copper Cross-Over


C:\>ping 2.2.2.2
Pinging 2.2.2.2 with 32 bytes of data:
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Ping statistics for 2.2.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Copper Cross-Over
Toggle PD


Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\ЛР\ЛР 13.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
13:26:30
Gig0/1
Gig0/2
3560-24PS
Multilayer Switch
Fa0/1
Fa0/2
Fa0/3
Fa0
PC-PT
PC0
PC1
PC2
PC3
PC4
PC5
PC6
VLAN2
VLAN3
Time: 00:26:49
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Copper Cross-Over
ENG
21:58
11.09.2022

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#
```

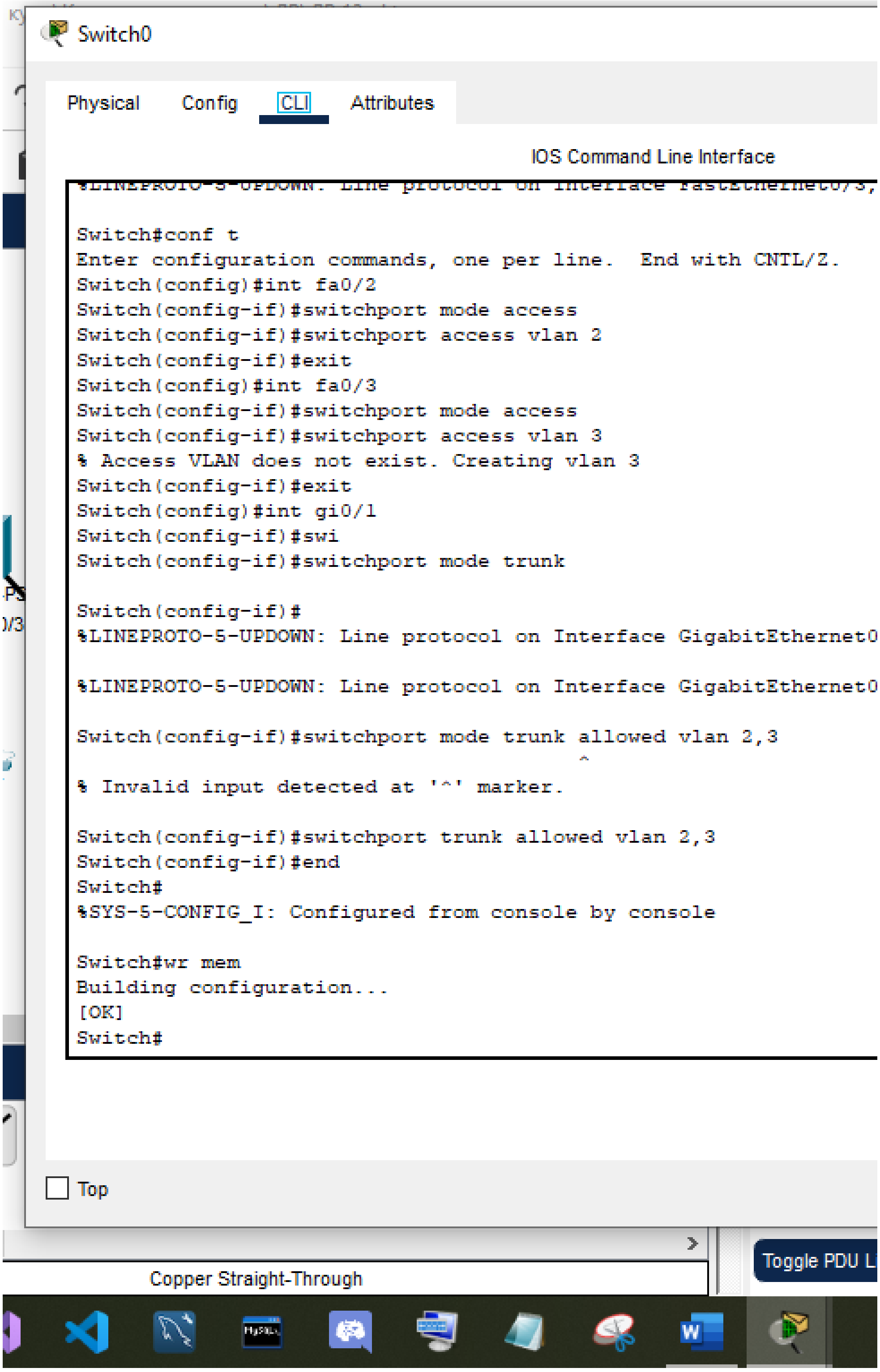
Top

Copper Straight-Through

Toggle PDU

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#
```

Top


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#int gi0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0
Switch(config-if)#switchport mode trunk allowed vlan 2,3
^
% Invalid input detected at '^' marker.
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#wr mem
Building configuration...
[OK]
Switch#
Top
Copper Straight-Through
Toggle PDU L

Switch1

Physical Config CLI Attributes

IOS Command

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

Top

Multilayer Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int gi0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk enca
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#swi
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#int vlan 2
Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 3
Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

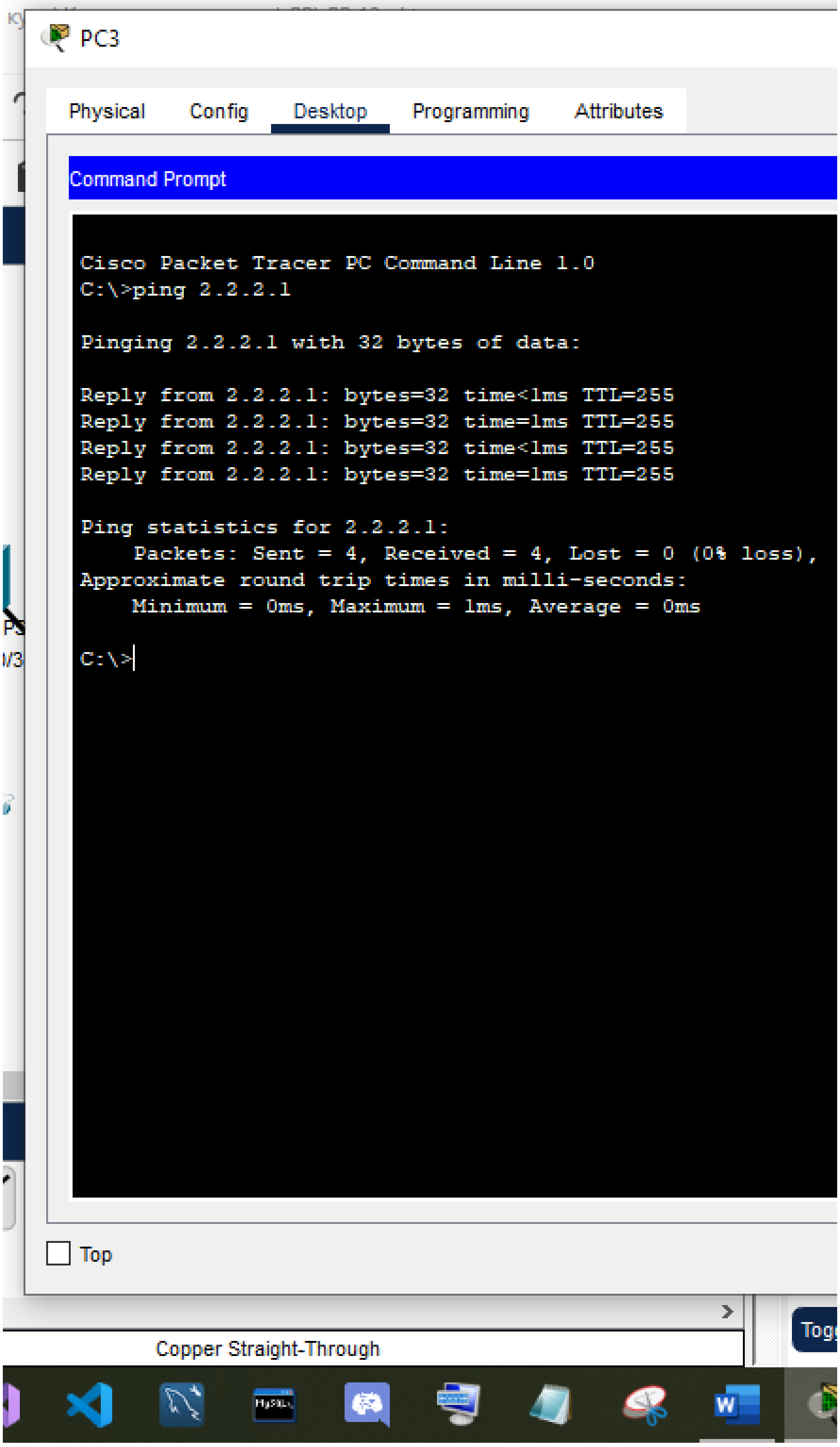
Top

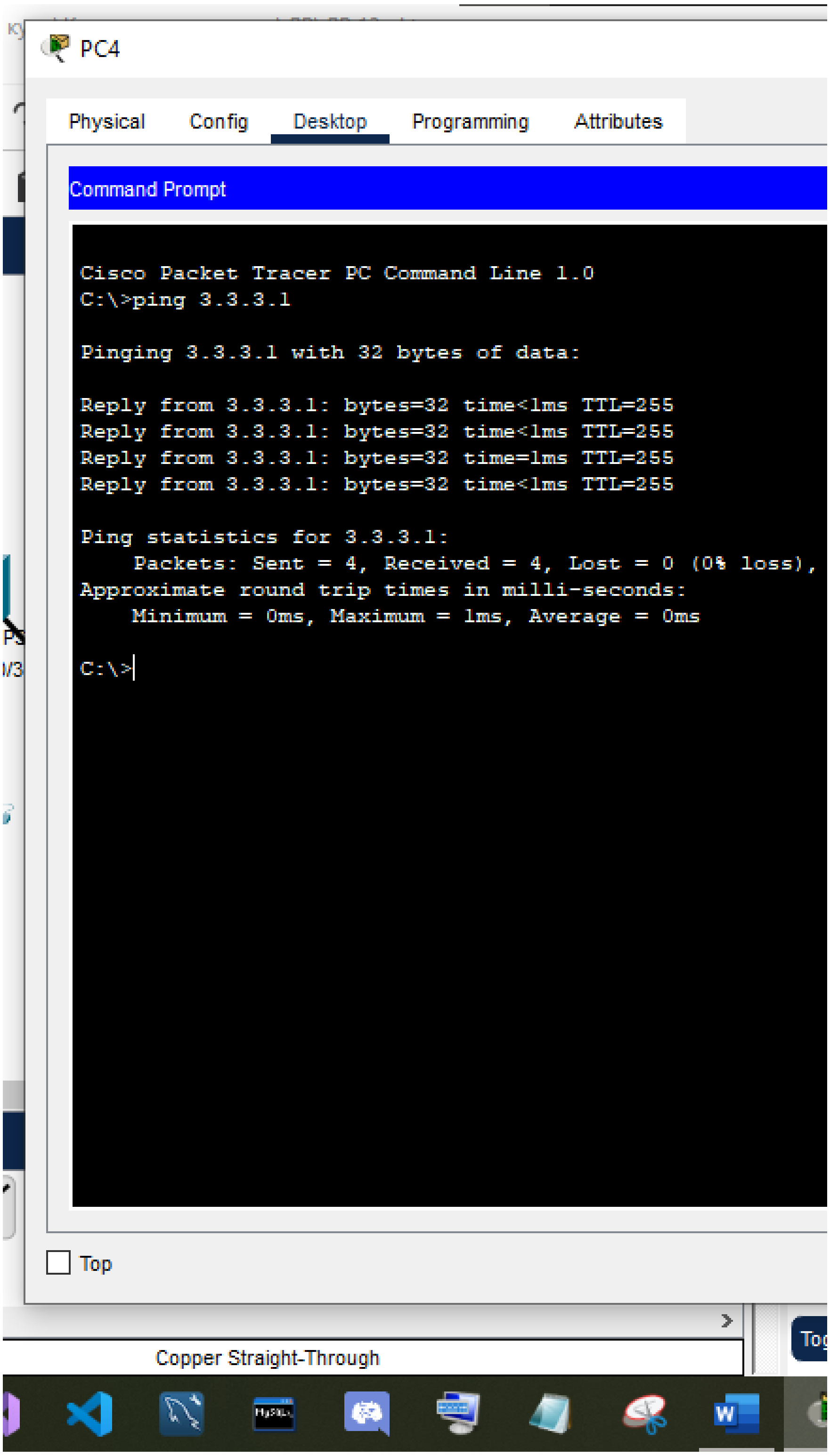
Multilayer Switch1

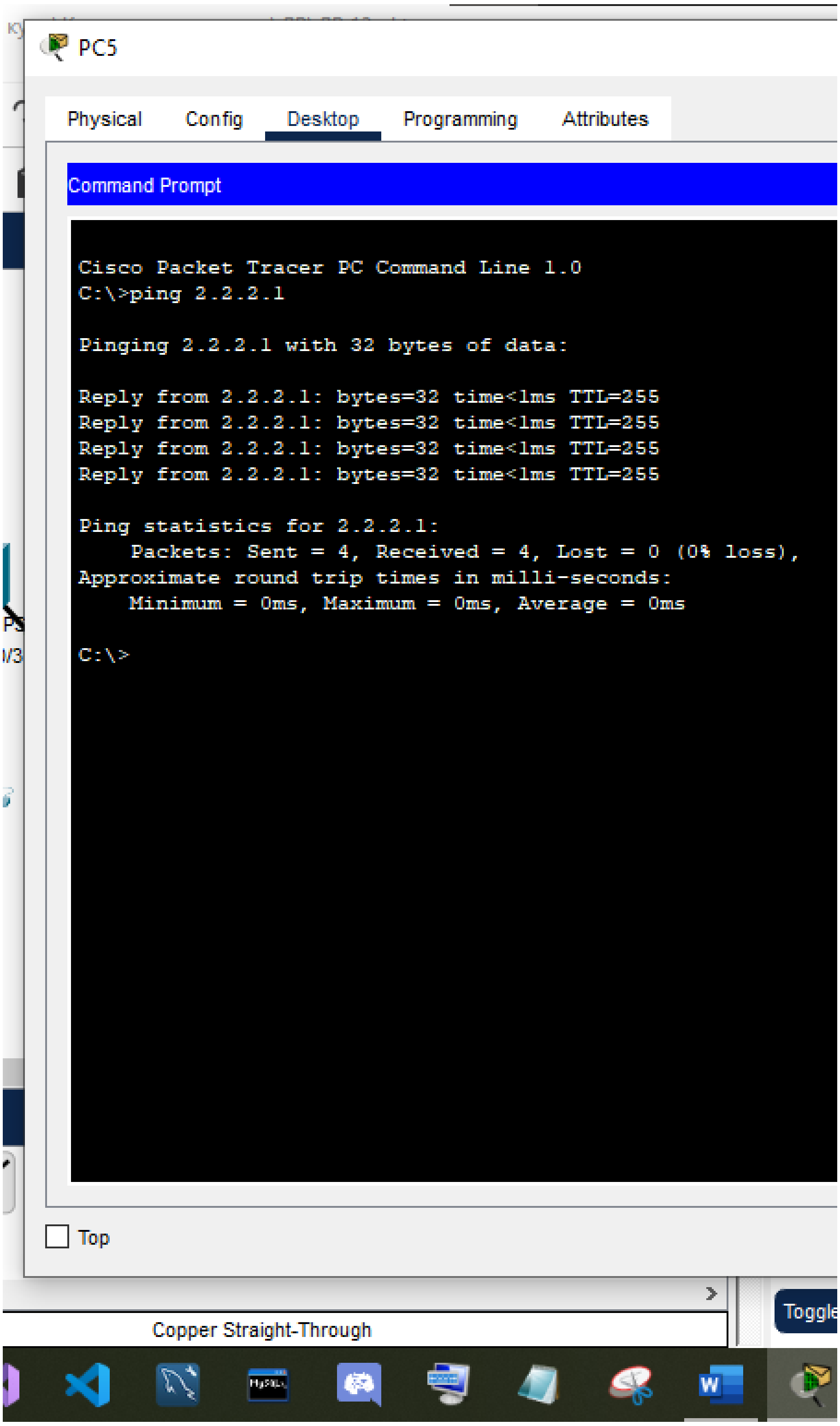
Physical Config CLI Attributes

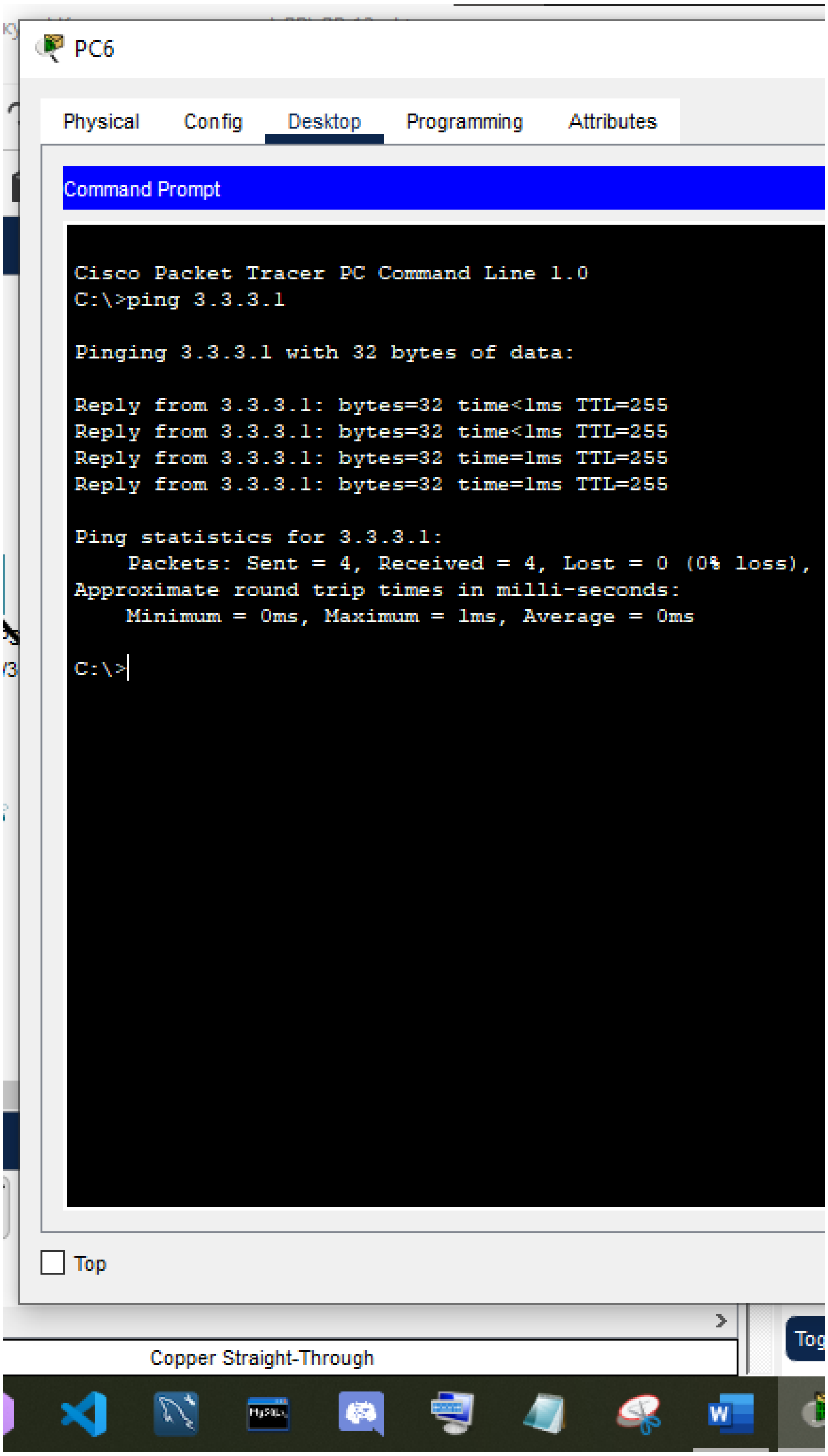
```
Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 3
Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#
%LINK-5-CHANGED: Interface Vlan3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3, changed state to up

Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

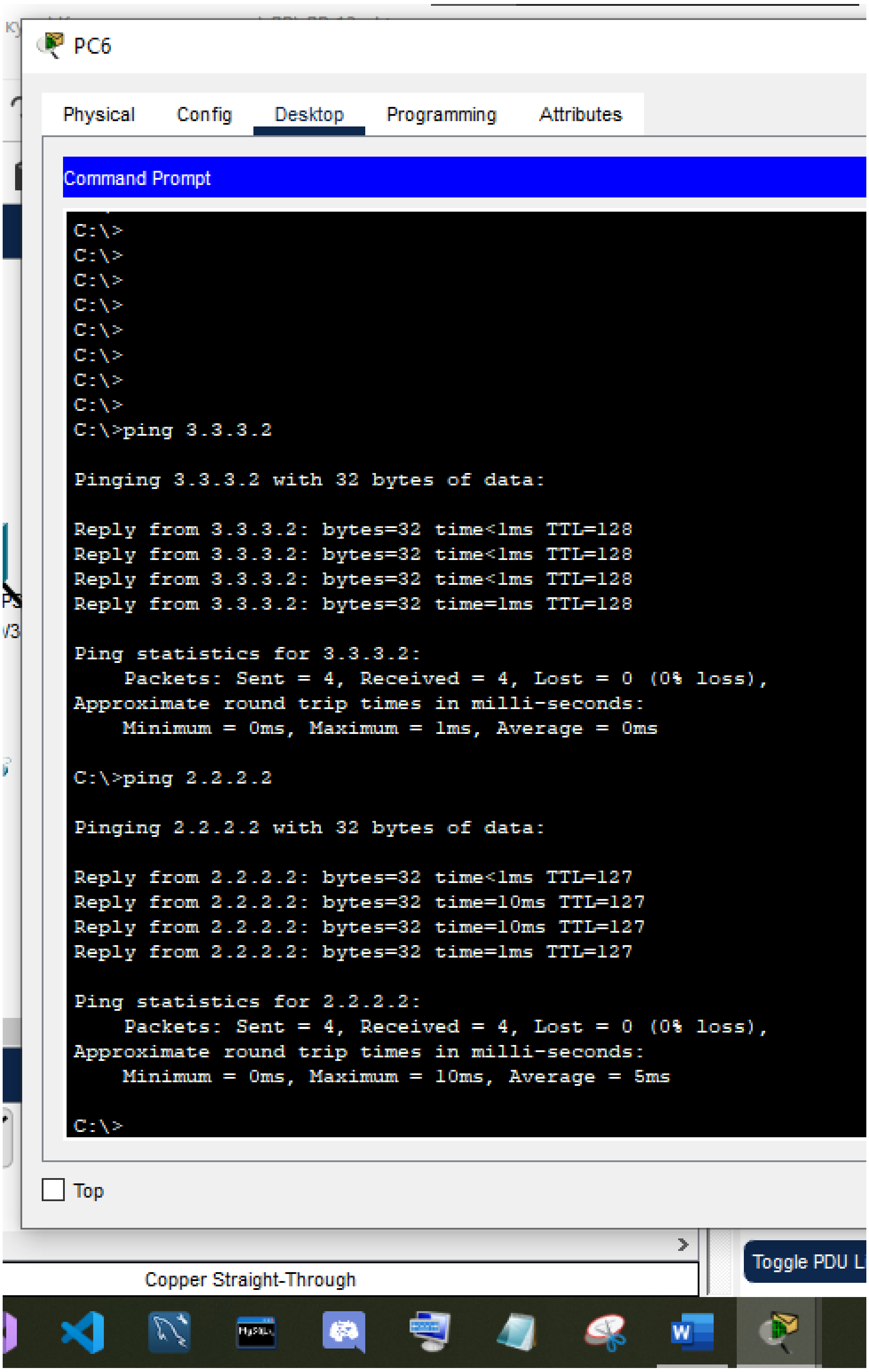
Top

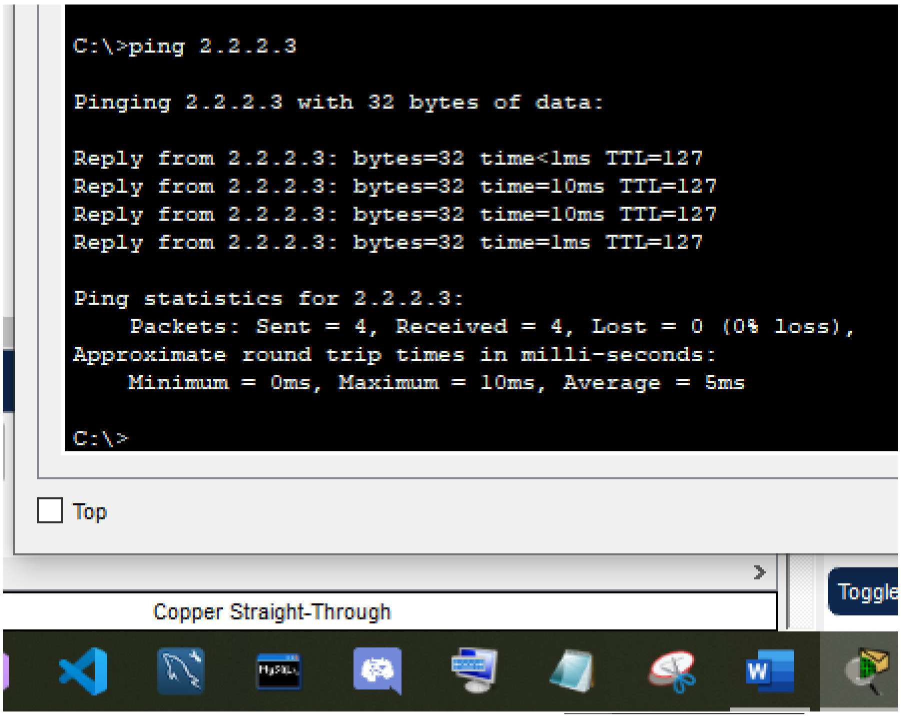

PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2.2.2.1
Pinging 2.2.2.1 with 32 bytes of data:
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time=1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time=1ms TTL=255
Ping statistics for 2.2.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Copper Straight-Through

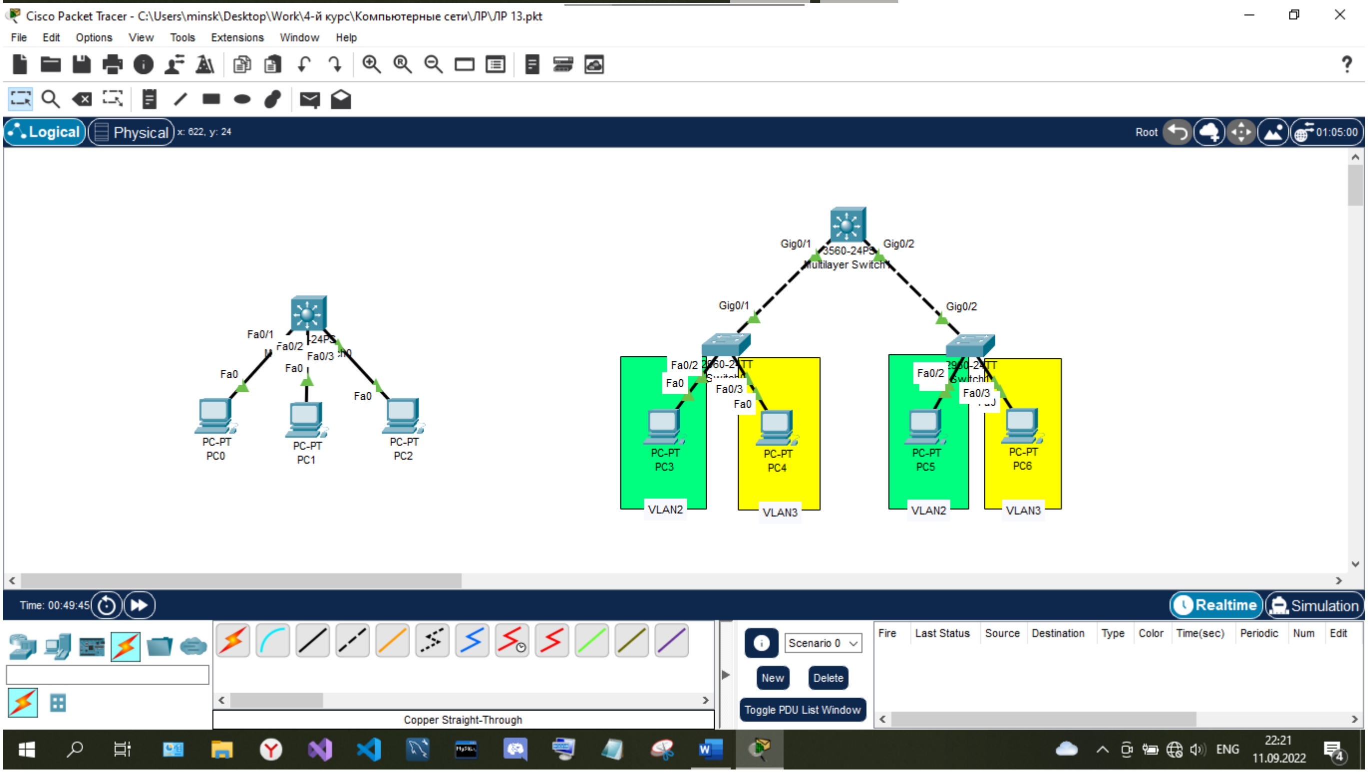

PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.3.3.1
Pinging 3.3.3.1 with 32 bytes of data:
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Ping statistics for 3.3.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Copper Straight-Through


PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2.2.2.1
Pinging 2.2.2.1 with 32 bytes of data:
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 2.2.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Copper Straight-Through


PC6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.3.3.1
Pinging 3.3.3.1 with 32 bytes of data:
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255
Ping statistics for 3.3.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Copper Straight-Through


PC6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 3.3.3.2
Pinging 3.3.3.2 with 32 bytes of data:
Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time=1ms TTL=128
Ping statistics for 3.3.3.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2.2.2.2
Pinging 2.2.2.2 with 32 bytes of data:
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Reply from 2.2.2.2: bytes=32 time=10ms TTL=127
Reply from 2.2.2.2: bytes=32 time=10ms TTL=127
Reply from 2.2.2.2: bytes=32 time=1ms TTL=127
Ping statistics for 2.2.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 5ms
C:\>
Top
Copper Straight-Through
Toggle PDU L





Контрольные вопросы

1. Перечислите операторы фильтров и то, что они значат

==/eq - Равно

!=/ne -Не равно

</lt - Меньше чем

<=/le - Меньше или равно

>/gt - Больше чем

>=/ge - Больше или равно

2. Как проходит Фильтрация IPv6 протокола?

   ipv6 – показывает трафик IPv6

3. Как фильтровать по номеру порта?

   ftp.port==<значение>

Вывод: я научился анализировать сетевой трафик.